

❖ Continuous-Time Markov Chains

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- Continuous-Time Markov Chain
- Chapman–Kolmogorov Equations
- Holding times
- Alarm Clocks and Transition Rates
- Infinitesimal Generator
- Absorbing states
- Time Reversibility

A Continuous-Time Weather Model

Recall the three-state weather model introduced earlier, with state space

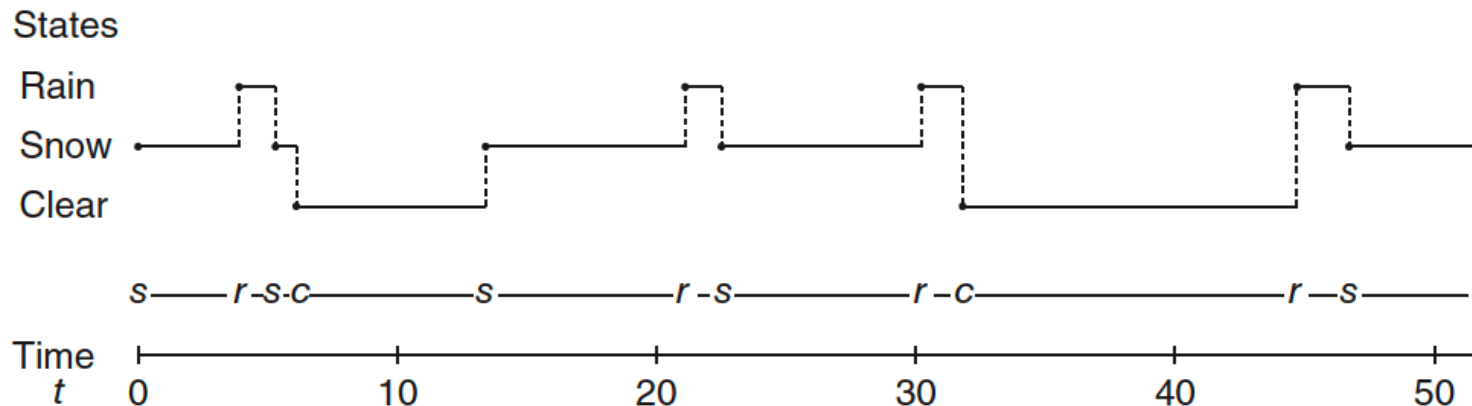
$$\Omega = \{\text{rain, snow, clear}\}.$$

Assume that:

- Rain lasts, on average, **3 hours**,
- Snow lasts, on average, **6 hours**, and
- Clear weather lasts, on average, **12 hours**.

Thus, the **holding** times in each state are exponentially distributed with parameters

$$(\lambda_r, \lambda_s, \lambda_c) = \left(\frac{1}{3}, \frac{1}{6}, \frac{1}{12} \right)$$



Transitions between weather states are governed by the stochastic matrix

$$\tilde{P} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{3}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{pmatrix}$$

Together with an initial distribution, sufficient to compute probabilities of the form

$$\mathbb{P}(X_{t_1} = i_1, \dots, X_{t_n} = i_n)$$

for any $n \geq 1$, states $i_1, \dots, i_n$, and times $t_1, \dots, t_n \geq 0$.

Markov Property

A continuous-time stochastic process $(X_t)_{t \geq 0}$ with discrete state space Ω is a **Continuous-Time Markov Chain** if

$$\mathbb{P}(X_{t+s} = j \mid X_s = i, X_u = x_u, 0 \leq u < s) = \mathbb{P}(X_{t+s} = j \mid X_s = i),$$

for all $s, t \geq 0$ and states $i, j \in \Omega$.

If these probabilities do not depend on s , the chain is **time-homogeneous**, i.e.,

$$\mathbb{P}(X_{t+s} = j \mid X_s = i) = \mathbb{P}(X_t = j \mid X_0 = i) \text{ for } s \geq 0$$

Transition Function

For a time-homogeneous CTMC $(X_t)_{t \geq 0}$, define the **transition function**

$$P_{ij}(t) := P(X_t = j \mid X_0 = i),$$

and collect these probabilities into a matrix

$$P(t) := (P_{ij}(t))$$

Chapman–Kolmogorov Equations

For continuous-time Markov process $(X_t)_{t \geq 0}$ with transition function $P(t)$,

The transition function satisfies

$$P(s + t) = P(s)P(t) \quad \text{for } s, t \geq 0$$

That is

$$P_{ij}(s + t) = \sum_k P_{ik}(s)P_{kj}(t)$$

for $s, t \geq 0$ and states i, j .

$$P_{ij}(s+t) = P(X_{s+t} = j \mid X_0 = i)$$

$$= \sum_k P(X_{s+t} = j \mid X_s = k, X_0 = i) P(X_s = k \mid X_0 = i)$$

$$= \sum_k P(X_{s+t} = j \mid X_s = k) P(X_s = k \mid X_0 = i)$$

$$= \sum_k P(X_t = j \mid X_0 = k) P(X_s = k \mid X_0 = i)$$

$$= \sum_k P_{ik}(s) P_{kj}(t)$$

Example: Poisson Process as a CTMC

Let $(N_t)_{t \geq 0}$ be a Poisson process with rate λ .

- State space: $\Omega = \{0, 1, 2, \dots\}$
- Transitions are only upward by 1
- Holding times are exponential with parameter λ

For $j \geq i$, $P_{ij}(t) = P(N_{t+s} = j \mid N_s = i)$

$$= \frac{P(N_{t+s} = j, N_s = i)}{P(N_s = i)} = \frac{P(N_{t+s} - N_s = j - i, N_s = i)}{P(N_s = i)}$$

$$= P(N_{t+s} - N_s = j - i)$$

$$= P(N_t = j - i) = \frac{e^{-\lambda t} (\lambda t)^{j-i}}{(j-i)!}$$

The transition matrix has the form:

$$P(t) = \begin{pmatrix} e^{-\lambda t} & \lambda t e^{-\lambda t} & \frac{(\lambda t)^2}{2!} e^{-\lambda t} & \dots \\ 0 & e^{-\lambda t} & \lambda t e^{-\lambda t} & \dots \\ 0 & 0 & e^{-\lambda t} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Holding Times

Let T_i denote the **holding time** at state i , that is, the length of time that a continuous-time Markov chain started in state i remains in state i before transitioning to a different state. Then

$$T_i \sim \text{Exponential}(q_i),$$

where $q_i > 0$ is the *exit rate* from state i .

$$\begin{aligned} P(T_i > s + t \mid X_0 = i) &= P(T_i > s + t \mid T_i > s, X_0 = i)P(T_i > s \mid X_0 = i) \\ &= P(T_i > s + t \mid X_u = i, \text{ for } 0 \leq u \leq i)P(T_i > s \mid X_0 = i) \\ &= P(T_i > s + t \mid X_s = i)P(T_i > s \mid X_0 = i) \\ &= P(T_i > t \mid X_0 = i)P(T_i > s \mid X_0 = i) \end{aligned}$$

For chain started in i , the event $\{T_i > s\} = \{X_u = i, \text{ for } 0 \leq u \leq s\}$

Embedded Chains

If we ignore the timing and only record the sequence of visited states

$$Y_0, Y_1, Y_2, \dots,$$

this defines a **discrete-time Markov chain**, called the **embedded chain**.

Let

$$\tilde{P}_{ij} = P(Y_{n+1} = j | Y_n = i)$$

then $\tilde{P}$ is a stochastic matrix with zeros on the diagonal.

Example (Poisson Process)

For a Poisson process with parameter λ , the holding time parameters are constant, i.e., $q_i = \lambda$, for $i = 0, 1, 2, \dots$

$$\tilde{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ \vdots \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & 0 & \dots \\ 0 & 0 & 0 & 1 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & \dots \\ 0 & 0 & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}.$$

□ Alarm Clocks and Transition Rates

Transition Rates

An equivalent way to define a CTMC is via **transition rates** q_{ij} for $i \neq j$

Alarm Clock

Associate with each possible transition (i, j) an **independent alarm clock**

When the chain enters state i , independent exponential clocks are started:

- One clock for each possible transition $i \rightarrow j$
- Clock (i, j) rings after an exponential time with rate q_{ij}

The first clock to ring determines the next state.

Holding time in state i :

Consider the process started in state i . The time until the next transition is the **minimum** of independent exponential random variables with parameters $q_{i1}, q_{i2}, \dots$

The minimum of independent exponentials is itself exponential, with parameter equal to the **sum** of the individual rates

$$T_i \sim \text{Exponential}\left(\sum_k q_{ik}\right) \qquad q_i = \sum_k q_{ik}$$

The parameter q_i represents the **total rate of departure** from state i .

Embedded chain transition probabilities:

$$P(\text{next state is } j \mid X \text{ leaves } i) = p_{ij} = \frac{q_{ij}}{q_i} = \frac{q_{ij}}{\sum_k q_{ik}}$$

Example.

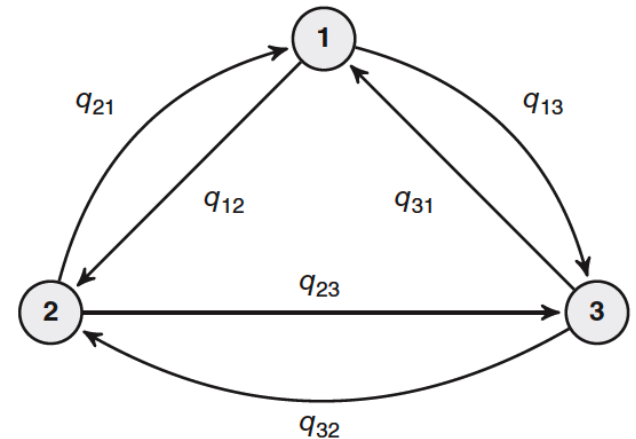
The general three-state continuous-time Markov chain is described by the transition graph

In terms of the transition rates, holding time parameters are

$$(q_1, q_2, q_3) = (q_{12} + q_{13}, q_{21} + q_{23}, q_{31} + q_{32}),$$

with embedded chain transition matrix

$$\tilde{P} = \begin{pmatrix} 0 & \frac{q_{12}}{q_1} & \frac{q_{13}}{q_1} \\ \frac{q_{21}}{q_2} & 0 & \frac{q_{23}}{q_2} \\ \frac{q_{31}}{q_3} & \frac{q_{32}}{q_3} & 0 \end{pmatrix}$$



Example: Registration Line (Queueing Model)

A registrar's office forms a line for students needing help.

- **Arrivals:** Students arrive according to a Poisson process at rate

$$\lambda = \frac{1}{4} \text{ per minute (one student every 4 minutes).}$$

- **Service:** The registrar serves a student in an exponential time with rate

$$\mu = \frac{1}{5} \text{ per minute (one student every 5 minutes).}$$

- **Capacity:** The line can hold at most 4 students. If a student arrives when the line already has 4 students, the student leaves and is not counted.

Let X_t be the number of students in line at time t . Then $(X_t)_{t \geq 0}$ is a CTMC on the state space $\{0, 1, 2, 3, 4\}$.

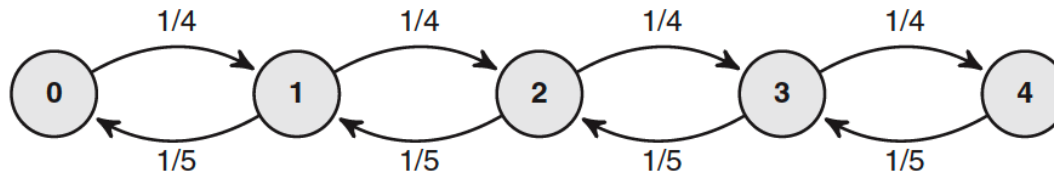
- Draw the transition-rate picture (rates on arrows).
- Find the holding-time parameters $q_0, \dots, q_4$.
- Find the embedded chain transition matrix $\tilde{P}$

1: Identify possible jumps and their rates q_{ij}

From a state i , there are two possible “clocks”:

- **Arrival clock** with rate $\lambda = \frac{1}{4}$, which increases the line by 1 (if not full).
- **Service clock** with rate $\mu = \frac{1}{5}$, which decreases the line by 1 (if not empty)

So the transition rates q_{ij} are:



2: Holding-time parameters q_i

$$q_i = \sum_k q_{ik}$$

$$q_0 = \frac{1}{4}.$$

$$q_1 = q_2 = q_3 = \frac{1}{4} + \frac{1}{5} = \frac{9}{20}.$$

$$q_4 = \frac{1}{5}.$$

3. Embedded chain transition matrix $\tilde{P}$

The embedded chain records only the next state visited after a jump. Its transition probabilities are

$$p_{ij} = \frac{q_{ij}}{q_i}$$

For example,

$$p_{0,1} = 1$$

$$p_{1,2} = \frac{5}{9}$$

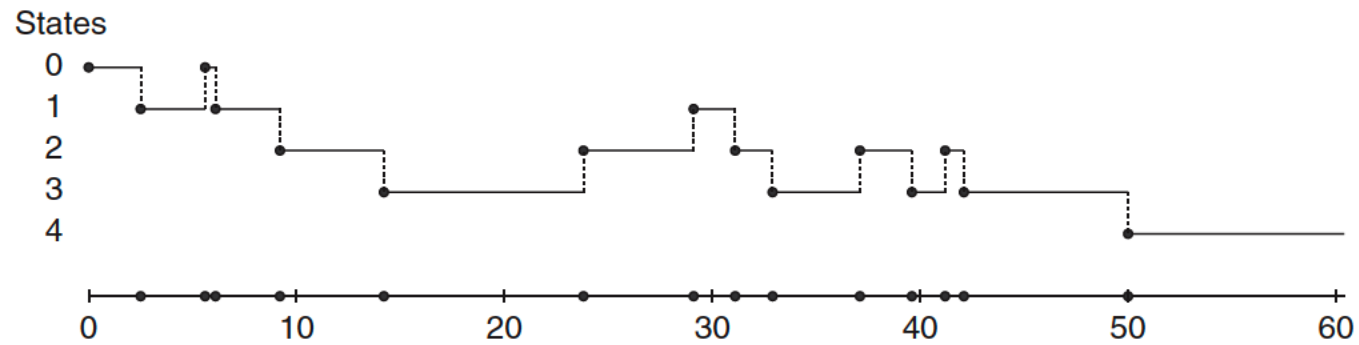
$$p_{1,0} = \frac{4}{9}$$

$$\tilde{P} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{4}{9} & 0 & \frac{5}{9} & 0 & 0 \\ 0 & \frac{4}{9} & 0 & \frac{5}{9} & 0 \\ 0 & 0 & \frac{4}{9} & 0 & \frac{5}{9} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Meaning:

$$P(\text{next state} = j \mid \text{current state} = i) = p_{ij} = \frac{q_{ij}}{q_i}$$

Each row answers exactly: “Given I just left state i , where do I go next?”



Instantaneous Rates q_{ij} , Holding Times q_i , Transition Probabilities p_{ij}

$$q_{ij} = q_i p_{ij}$$

$$p_{ij} = \frac{q_{ij}}{q_i}$$

□ Infinitesimal Generator

Let $(X_t)_{t \geq 0}$ be a continuous-time Markov chain on a **countable** state space Ω .

The transition function:

$$P_{ij}(t) := P(X_t = j \mid X_0 = i),$$

In particular,

$$P_{ij}(0) := \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$$

The **infinitesimal generator** $Q = (q_{ij})_{i,j \in \Omega}$ is the matrix that captures the *instantaneous* transition behavior.

$$q_{ij} = \lim_{h \rightarrow 0^+} \frac{\mathbb{P}(X_{t+h} = j \mid X_t = i) - \mathbb{P}(X_t = j \mid X_t = i)}{h}$$

- For $i \neq j$

$$\begin{aligned}
 q_{ij} &= \lim_{h \rightarrow 0^+} \frac{\mathbb{P}(X_{t+h} = j \mid X_t = i) - \mathbb{P}(X_t = j \mid X_t = i)}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{\mathbb{P}(X_h = j \mid X_0 = i)}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{P_{ij}(h)}{h} \\
 &= P'_{ij}(0)
 \end{aligned}$$

- For $i = j$

$$\begin{aligned}
 q_{ii} &= \lim_{h \rightarrow 0^+} \frac{\mathbb{P}(X_{t+h} = i \mid X_t = i) - \mathbb{P}(X_t = i \mid X_t = i)}{h} = \lim_{h \rightarrow 0^+} \frac{P_{ii}(h) - 1}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{-\sum_{j \neq i} P_{ij}(h)}{h} = -\sum_{j \neq i} q_{ij}
 \end{aligned}$$

Property: (generator matrix)

- Off-diagonals are nonnegative: $q_{ij} \geq 0$ for $i \neq j$.
- Rows sum to zero:

$$\sum_{j \in S} q_{ij} = 0 \quad \implies \quad q_{ii} = - \sum_{j \neq i} q_{ij} \leq 0.$$

Define the total rate out of state i as $q_i := -q_{ii} = \sum_k q_{ik}$

Example: Registration line(finite-capacity birth–death)

$$Q = \begin{pmatrix} -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{5} & -\frac{9}{20} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{5} & -\frac{9}{20} & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{5} & -\frac{9}{20} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{5} & -\frac{1}{5} \end{pmatrix}$$

Example Poisson process generator (pure birth)

For a Poisson process $(N_t)_{t \geq 0}$ with parameter λ , the holding time parameters are constant, i.e., $q_i = \lambda$, for $i = 0, 1, 2, \dots$

Only one possible jump: $n \rightarrow n + 1$

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & \cdots \\ 0 & -\lambda & \lambda & 0 & 0 & \cdots \\ 0 & 0 & -\lambda & \lambda & 0 & \cdots \\ 0 & 0 & 0 & -\lambda & \lambda & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{pmatrix}$$

Kolmogorov equations

A continuous-time Markov chain with transition function $P(t)$ and infinitesimal generator Q satisfies

- Forward equation

$$\frac{d}{dt}P(t) = P(t)Q \qquad P(0) = I$$

- Backward equation

$$\frac{d}{dt}P(t) = QP(t) \qquad P(0) = I$$

Finite-state case

If S is finite, then solution to the Kolmogorov equation gives

$$P(t) = e^{tQ} = \sum_{n=0}^{\infty} \frac{(tQ)^n}{n!}$$

Limiting Distribution

Let $(X_t)_{t \geq 0}$ be a continuous-time Markov chain on a **countable** state space Ω , with transition function:

$$P_{ij}(t) := P(X_t = j \mid X_0 = i),$$

A probability vector $\pi = (\pi_j)$ is a **limiting distribution** of a CTMC if

$$\lim_{t \rightarrow \infty} P_{ij}(t) = \pi_j \quad \text{for all } i, j$$

Regardless of where the chain starts, its distribution converges to π .

Stationary Distribution

A probability vector π is **stationary** if

$$\pi = \pi P(t) \quad \text{for any } t \geq 0$$

Equivalently, for all states j ,

$$\pi_j = \sum_i \pi_i P_{ij}(t)$$

Every limiting distribution of a continuous-time Markov chain is stationary.

The converse is **not always true** (depending on irreducibility.)

A CTMC is **irreducible** if for all i, j , if $P_{ij}(t) > 0$ for some $t > 0$.

Stationary Distribution and Generator

A probability vector π is stationary **if and only if**

$$\pi Q = 0$$

$$\sum_j \pi_j = 1$$

Fundamental Limit Theorem for Finite CTMC

Theorem

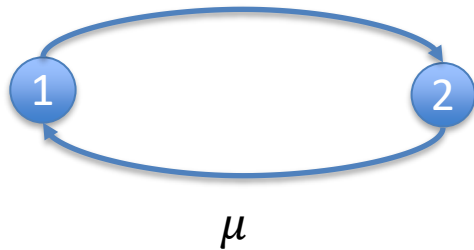
Let $(X_t)_{t \geq 0}$ be a finite, irreducible, continuous-time Markov chain.

- There exists a **unique stationary distribution** π .
- π is the **limiting distribution**:
$$\lim_{t \rightarrow \infty} P_{ij}(t) = \pi_j$$

This is the continuous-time analogue of the discrete-time ergodic theorem.

π_i is the long-run fraction of time spent in state i .

Example: (two state CTMC) λ



The generator is

$$Q = \begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix}$$

Solve $\pi Q = 0$

Example (Eat, play, sleep) Jesse is a newborn baby who is always in one of three states: eat, play, and sleep. He eats on average for 30 minutes at a time; plays on average for 1 hour; and sleeps for about 3 hours. After eating, there is a 50–50 chance he will sleep or play. After playing, there is a 50–50 chance he will eat or sleep. And after sleeping, he always plays. Jesse’s life is governed by a continuous-time Markov chain. What proportion of the day does Jesse sleep?

The holding time parameters for the three-state chain (in hour units) are

$$(q_e, q_p, q_s) = \left(2, 1, \frac{1}{3}\right).$$

The embedded chain transition probabilities are

$$\tilde{P} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix}$$

Solve Q using $\frac{q_{ij}}{q_i} = p_{ij}$

Absorbing Continuous-Time Markov Chains

A state i of a continuous-time Markov chain is **absorbing** if, once entered, the process never leaves:

$$q_i = 0.$$

An **absorbing Markov chain** is one in which there is at least one absorbing state.

Write the generator Q in canonical block matrix form, (absorbing states first)

$$Q = \begin{bmatrix} 0 & 0 \\ * & V \end{bmatrix}$$

For each transient state $i \in T$, define

$$a_i = E_i[\text{time until absorption}].$$

Theorem (Mean Time Until Absorption)

Let the fundamental matrix $F = -V^{-1}$. Then

- F_{ij} is the expected time spent in state j before absorption, given the chain starts in transient state i .
- The mean time until absorption starting from i is

$$a_i = \sum_{j \in T} F_{ij}$$

Discrete-time MC

$$P = \begin{pmatrix} I & 0 \\ R & Q \end{pmatrix}$$

Fundamental matrix $N = (I - Q)^{-1}$

Expected visits

Counts steps

Continuous-time MC

$$Q = \begin{pmatrix} 0 & 0 \\ * & V \end{pmatrix}$$

Fundamental matrix $F = -V^{-1}$

Expected **time**

Measures time

Example. Multistate Models (Disease Progression)

Multistate continuous-time Markov chains are widely used in **medical survival analysis** to model disease progression:

- Patients move through **increasingly severe disease stages**
- Transitions occur randomly in continuous time
- Eventually, the process reaches a **terminal absorbing state** (e.g. death)

We consider a three-state CTMC:

State	Meaning	Type
1	Cirrhosis	Transient
2	Liver cancer (HCC)	Transient
3	Death	Absorbing

The infinitesimal generator is

$$Q = \begin{pmatrix} -(q_{12} + q_{13}) & q_{12} & q_{13} \\ 0 & -q_{23} & q_{23} \\ 0 & 0 & 0 \end{pmatrix}$$

The transient–transient block of Q is

$$V = \begin{pmatrix} -(q_{12} + q_{13}) & q_{12} \\ 0 & -q_{23} \end{pmatrix}$$

For absorbing CTMCs, the **fundamental matrix** is

$$F = \begin{pmatrix} \frac{1}{q_{12} + q_{13}} & \frac{q_{12}}{q_{23}(q_{12} + q_{13})} \\ 0 & \frac{1}{q_{23}} \end{pmatrix}$$

$$F_{ij} = E_i[\text{Starting in } i \text{ total time spent in state } j \text{ before absorption}].$$

he mean time to absorption starting from transient state i is $a_i = \sum_{j \in T} F_{ij}$

$$a_1 = \frac{1}{q_{12} + q_{13}} + \frac{q_{12}}{q_{23}(q_{12} + q_{13})} \qquad a_2 = \frac{1}{q_{23}}$$

Data-based parameter estimates

From 1,925 patients, Bartolomeo et al. estimate:

$$\hat{q}_{12} = 0.0151, \quad \hat{q}_{13} = 0.0071, \quad \hat{q}_{23} = 0.0284 \quad (\text{per month}).$$

Plugging into F :

$$\hat{F} = \begin{pmatrix} 45.05 & 23.95 \\ 0 & 35.21 \end{pmatrix}$$

For a patient starting in **cirrhosis**:

$$\text{Mean time to death} = 45.05 + 23.95 = 69 \text{ months}$$

Time Reversibility

Definition. A continuous-time Markov chain with generator Q and stationary distribution π is **time reversible** if

$$\pi_i q_{ij} = \pi_j q_{ji} \text{ for all } i, j$$

This condition is called **local balance** or detailed balance.

Theorem. If a probability vector λ satisfies

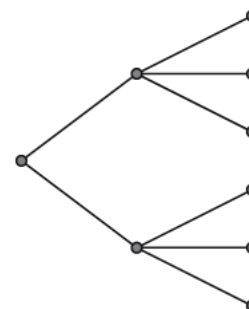
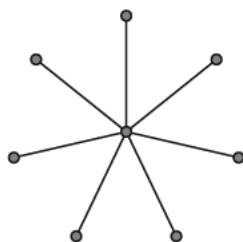
$$\lambda_i q_{ij} = \lambda_j q_{ji} \text{ for all } i, j$$

Then, the CTMC is time reversible, and λ is the unique stationary distribution.

Markov Processes on Trees

A **tree** is an undirected graph that is connected, and contains **no cycles**.

That is, between any two vertices there is exactly one simple path.



Given a CTMC with generator $Q = (q_{ij})$, define its undirected transition graph:

- vertices = states
- an (undirected) edge between i and j if

$$q_{ij} > 0 \text{ or } q_{ji} > 0.$$

Theorem (Markov Processes on Trees are Time Reversible)

If the transition graph of an irreducible continuous-time Markov chain is a **tree**, then the process is **time reversible**.

Example. Let the generator be

$$Q = \begin{pmatrix} -a & a & 0 \\ b & -(b+c) & c \\ 0 & d & -d \end{pmatrix}, \quad a, b, c, d > 0.$$

Example 2

$$Q = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

This is irreducible. The stationary distribution is uniform:

$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

But detailed balance fails: for example,

$$\pi_1 q_{12} =$$

$$\pi_2 q_{21} =$$

Example

$$Q = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

Because Q is symmetric, the uniform distribution is stationary:

$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

Check

$$\pi_i q_{ij} = \pi_j q_{ji} \text{ for all } i, j$$

The chain is **time reversible**.

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross